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Preventive Medicine

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Review Article

Physical activity and sedentary time during childcare outdoor play sessions: A systematic review and meta-analysis



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ARTICLE INFO

Keywords:

Physical activity
Sedentary time
Childcare
Young children
Outdoor play
Systematic review
Meta-analysis

ABSTRACT

Outdoor playtime has been highly correlated with moderate-to vigorous-intensity physical activity (MVPA), while also being associated with decreased sedentary time. This systematic review and meta-analysis sought to examine the physical activity levels and sedentary time of young children (2–5 years) during outdoor play periods at centre-based childcare. Eight online databases were searched for peer-reviewed, English-language, original research. Two reviewers independently extracted data (study design, participants, method of measurement, and physical activity and/or sedentary time of participants). Time spent in outdoor activity intensities were converted to percentage of time, and averaged across the two measurement types: accelerometry and observational. Of the 26 included articles, 11 used accelerometry, 13 employed direct observation, and 2 adopted a combination of both measurement methods. Average percentage of time engaged in MVPA and sedentary pursuits ranged from 6.7% to 43% and 23.2% to 63.5% by accelerometry, and 2% to 53.9% and 46.1% to 80.2% by observation, respectively. Total physical activity (only measured by accelerometry) ranged from 23.2% to 63.5% of outdoor playtime. Meta-analysis of 11 accelerometer studies found that children spent a mean (95% CI) of: 13.99% (10.35–17.63) of outdoor playtime in MVPA, 43.77% (32.95–54.58) in TPA, and 52.81% (44.1–61.51) in sedentary time. Overall, young children have the potential to be highly active during outdoor play sessions at centre-based childcare; however, many children still engage in substantial amounts of sedentary time. Future research should explore the frequency and/or duration of outdoor time which best support young children's physical activity and limits sedentary time in childcare.

1. Introduction

The low levels of physical activity and markedly high levels of sedentary time among young children in childcare is concerning (Vanderloo et al., 2015). Researchers have determined that preschoolers in this environment accumulate a mere 1.54 min/h of moderate-to vigorous-intensity physical activity (MVPA) (Vanderloo et al., 2015), and spend the majority of their time (up to 40.64 min/h) being sedentary (Vanderloo et al., 2014; Reilly, 2010). Unfortunately, participation in physical activity negatively correlates with age; (Garriguet et al., 2016) young children's activity levels decrease by almost 50% between the ages of 3 and 5 years (Taylor et al., 2013). Canada, Australia, and the United Kingdom recommend that young children 2 to

4 years engage in at least 180 min of total physical activity (TPA) per day (Tremblay et al., 2017; Okely et al., 2017; Department of Health, 2011), while Canada and Australia have recently updated their guidelines recommending that children 3 years and older accumulate a minimum of 60 min of this time in energetic (i.e., MVPA) play (Tremblay et al., 2017; Okely et al., 2017). Acknowledging that activity behaviours established in early childhood have been shown to track across the lifespan (Cleveland et al., 2008), there is a need to promote healthy activity habits from a young age.

Not participating in physical activity in early childhood negatively affects healthy development (Tucker and Irwin, 2008). Physical activity is associated with a number of psychosocial, cognitive, and physical health benefits including increased mental wellness (Ahn and Fedewa,

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<https://doi.org/10.1016/j.ypmed.2017.12.022>

Received 7 June 2017; Received in revised form 19 December 2017; Accepted 22 December 2017

Available online 03 January 2018

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2011; Penedo and Dahn, 2005), enhanced socialization (Lees and Hopkins, 2013; Timmons et al., 2012), more favourable cardiometabolic health markers (Timmons et al., 2012), decreased insulin resistance (Bell et al., 2007), and greater academic achievement (Reed et al., 2010). Further, too much time spent being sedentary increases the likelihood of developing weight-related issues (childhood obesity is known to affect over 43 million children under the age of 5 worldwide) (World Health Organization, 2015), type 2 diabetes, and heart disease, which present additional health complications, including hyperlipidemia (Goel, 2015) and hypertension (Goel, 2015; Barton, 2012).

Centre-based childcare has been noted as a prime environment for targeting young children's health behaviours, as a significant portion of this population is enrolled in this type of childcare arrangement for approximately 30 h per week (Cleveland et al., 2008). Moreover, childcare centres have been identified as being a strong contributor to young children's physical activity levels (Vanderloo et al., 2014; Finn et al., 2002; Trost et al., 2010). With the childcare environment representing a good catchment area for this population, a number of physical activity interventions have transpired (Tucker and Irwin, 2008; Finn et al., 2002; Pate et al., 2008). Specifically, many interventions have used outdoor playtime as a mechanism to increase young children's physical activity levels (Brown et al., 2009a; Nicaise et al., 2011; Nicaise et al., 2012; Trost et al., 2008), as children are significantly more active when they are outdoors compared to indoors (Raustorp et al., 2012; Vanderloo et al., 2013). For example, Raustorp et al. (2012) reported that young children at childcare in the United States and Sweden spent 7.8% and 6.7% of their time outdoors in MVPA, compared to just 1.8% and 2.9% of their time in this behaviour indoors, respectively. Similarly, in Canada, Vanderloo et al. (2013) identified that preschoolers were actually 10 times more active outside than inside, with MVPA rates of 5.03 min/h and 0.54 min/h, respectively. Outdoors, children have the increased ability to engage in gross motor activity and master their emerging fundamental movement skills such as running, jumping, throwing, catching, and striking (Timmons et al., 2012). Children who engage in active outdoor play are able to explore their natural environment and learn to appreciate the outdoors (Tremblay et al., 2015), while also demonstrating resiliency, self-regulation, the ability to socialize with peers, and developing skills for dealing with stress later in life (Brussoni et al., 2012; Greever et al., 2015; Veitch et al., 2006). Highlighting the best available scientific evidence, an expert derived position statement on active outdoor play emphasized that the physical, mental, social, and environmental benefits of outdoor affordances greatly outnumber the perceived harms of injury and abduction, etc. (Tremblay et al., 2015).

Due to the large portion of time young children spend at childcare, it is an essential environment for this group to develop healthy lifestyle behaviours. While there is general consensus, via independent studies (Raustorp et al., 2012; Vanderloo et al., 2013), that children are more active outdoors than indoors, no synthesis has been completed to provide a comprehensive look at outdoor playtime among young children in centre-based childcare to date. Given the varied approaches in measurement and study design, and the inconsistencies in outdoor playtime allotment, a synthesis of outdoor physical activity and sedentary time in childcare centres is warranted. Therefore, the purpose of this systematic review and meta-analysis was to assess and synthesize young children's physical activity levels and sedentary time during outdoor play sessions at centre-based childcare. Highlighting the activity levels of young children in this specific setting has the potential to impact policy makers, early year's stakeholders, as well as early childhood educators in prioritizing outdoor time for optimal health in young children.

2. Methods

This review was registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration no.

Table 1
Sample Search Strategy (EMBASE).

#	Search term	Results	Search type
1	preschool child/	547,180	Advanced
2	"preschoolers".mp	5836	Advanced
3	toddler.mp.	7394	Advanced
4	toddlers.mp.	7518	Advanced
5	"early years".mp.	4177	Advanced
6	"early childhood".mp.	29,039	Advanced
7	"preschool-aged child".mp.	47	Advanced
8	"preschool-aged children".mp.	1616	Advanced
9	"young child".mp.	4684	Advanced
10	"young children".mp.	51,507	Advanced
11	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10	1,459,133	Advanced
12	childcare.mp OR exp child care/	64,155	Advanced
13	"childcare centre".mp.	31	Advanced
14	"centre-based childcare".mp.	17	Advanced
15	"centre-based childcare".mp.	18	Advanced
16	"centre-based childcare".mp.	18	Advanced
17	"centre-based childcare".mp.	17	Advanced
18	"day care".mp.	14,895	Advanced
19	"early learning centre".mp.	1	Advanced
20	"early learning centre".mp.	2	Advanced
21	physical activity.mp. OR physical activity/	155,492	Advanced
22	exercise.mp.	433,622	Advanced
23	movement.mp.	361,251	Advanced
24	"active play".mp.	220	Advanced
25	"locomotor activity".mp.	24,357	Advanced
26	"motor activity".mp.	54,511	Advanced
27	"physical exertion".mp.	2798	Advanced
28	"active movement".mp.	1274	Advanced
29	"outdoor play".mp.	217	Advanced
30	outdoor time.mp.	250	Advanced
31	recess.mp.	115	Advanced
32	"sedentary behaviour".mp.	1547	Advanced
33	sedentary lifestyle/or sedentary.mp	34,250	Advanced
34	Inactive.mp.	106,149	Advanced
35	stationary.mp.	57,673	Advanced
36	"physical inactivity".mp.	9087	Advanced
37	"sedentary activity".mp.	597	Advanced
38	12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20	170,150	Advanced
39	21 OR 22 OR 23 OR 23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 OR 32 OR 33 OR 34 OR 35 OR 36 OR 37	1,593,829	Advanced
40	11 AND 38 AND 39	802	Advanced

CRD42017062337), and adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement for systematic reviews (Liberati et al., 2009; Moher et al., 2015).

2.1. Search strategy

A comprehensive search strategy, designed by the Health Sciences Librarian, was implemented by the research team to assess the physical activity and sedentary time among young children during outdoor play sessions in centre-based childcare. Eight electronic databases (CINAHL, EMBASE, Medline, Physical Education Index, ProQuest, PsycInfo, SCOPUS, and Sport Discus) were searched for keywords such as "physical activity", "sedentary behaviours", "sedentary time", "preschoolers", "young children", "childcare", and "outdoor play". See Table 1 for a search strategy example. No date restrictions were applied, and limits were not placed on publication type to be inclusive during initial stages of the literature search. The final database search was run on February 10, 2017. All retrieved papers were exported into independent folders in Mendeley (version 1.17) referencing software. Papers from each database were then combined into one folder and duplicates were removed.

2.2. Eligibility criteria

In order to be included in the systematic review, papers had to: be original research; be published in English in a peer-reviewed journal; focus on typically developing (i.e., free from chronic diseases or developmental issues) young children enrolled in centre-based childcare (2 to 5 years); measure physical activity and/or sedentary time objectively (i.e., accelerometry or direct observation); focus on physical activity and/or sedentary time during an outdoor play session at centre-based childcare; and present results in percentage of time, or provide enough information (e.g., total outdoor time) to calculate percentage of time spent in different intensities of physical activity (i.e., MVPA and TPA). For inclusion in the meta-analysis, studies had to meet the above requirements and measure physical activity and/or sedentary time by accelerometer, provide the standard deviation (or enough information to calculate a standard deviation) for each/any activity intensity, and report a sample size. In instances where results were presented in figure or graphical format, or information to calculate percentage of time engaged in activity intensities was missing, authors were contacted to seek additional information.

A screening form was used by two independent reviewers to assess the title and abstracts of all potentially relevant articles. Once complete, the reviewers compared their results for confirmation and generated a list of articles to retrieve in full. Any discrepancies between the two reviewers resulted in the paper being included for full-text screening. The same two reviewers independently read each full-text article and determined their appropriateness for inclusion in this review. A third reviewer was brought in for consultation in instances of uncertainty, and a final set of articles was agreed upon.

To ensure all appropriate literature was captured, the reference lists of the studies identified for full-text screening were also examined. The 'ahead of print' sections of three highly-cited journals in the physical activity field (*Pediatric Exercise Science, Medicine and Science in Sports and Exercise*, and the *Journal of Physical Activity and Health*) were also searched to confirm that the most up-to-date studies were captured in this review.

2.3. Data extraction

Two researchers independently extracted the following information from each article: 1. study characteristics (i.e., authors, publication date, country, sample description, study design); 2. method of activity measurement (i.e., accelerometry, observational); and, 3. outcome variable (i.e., physical activity [MVPA, TPA] and/or sedentary time). Light physical activity (LPA) was not extracted as most of the included studies did not report this outcome variable. Depending on the method of activity measurement, additional information was extracted. Specifically, where accelerometry was used, average wear-time and monitoring time of accelerometers were recorded, as well as the epoch length and cut-points applied. For observational studies, length of observation as well as total number of observations were recorded. Additional studies ($n = 5$) were excluded from this review because adequate information was not provided for interpreting the data (i.e., total outdoor time measured was not reported).

2.4. Quality assessment and risk of bias

In order to assess the quality of the studies included in this review, a modified version of the Downs and Black checklist (questions 1–3, 6, 7, 10–12, 18 and 20) was used (Downs and Black, 1998), in line with previous approaches using this tool (Duch et al., 2013; Vanderloo, 2014). The original Downs and Black checklist has five sections including questions about: study quality, external validity, study bias, confounding and selection bias, and power of the study. As outcomes of the current study were not based on interventions, specific questions were negated from the checklist. All articles were assessed for quality

by two independent reviewers. Each article was assigned to one of three categories based on its score from the modified checklist (i.e., *low* = 0–3, *medium* = 4–6, *high* = 7–10). Due to the limited available research examining outdoor playtime in childcare, the desire for a comprehensive picture of activity during outdoor play, and for the purpose of this review, all studies were included regardless of their quality score.

2.5. Data synthesis and analysis

To facilitate comparisons and data synthesis, studies were separated based on measurement method (i.e., accelerometry, observational). For accelerometry studies, physical activity was categorized as MVPA and TPA, while sedentary time was considered independently. If not already provided, results for accelerometry and observational studies were converted to percentage of time for each activity intensity to allow for comparability between studies. If data for a particular intensity level was not provided, simple calculations were completed using the data provided in-text (i.e., adding time spent in moderate-intensity physical activity and vigorous-intensity physical activity together to create MVPA).

For observational studies that used the Observational System for Recording Physical Activity in Children–Preschool (OSRAC-P) or the System for Observing Fitness Instruction Time – Preschoolers (SOFIT-P), a rating scale from 1 to 5 was applied (1 = stationary with no movement, 2 = stationary with limb movement, 3 = slow/easy translocation, 4 = medium/moderate translocation, 5 = very fast/strenuous translocation) (Brown et al., 2006). The mean activity levels were transferred to intensities, with levels 1 and 2 representing sedentary time, and levels 4 and 5 were combined to form MVPA (Kahan et al., 2013). Aggregating levels 4 and 5 has been a common procedure in many physical activity studies for assessing and reporting children's non-sedentary activity (Pate et al., 2008; McKenzie et al., 1997). The study which used the System for Observing Play and Leisure Activity in Youth (SOPLAY) quantified activity on a scale from 1 to 3, with 1 representing sedentary behaviour, 2 indicating moderate intensity activity, and 3 signifying vigorous activity. The mean percentage of time spent in MVPA was calculated by summing the time spent in levels 2 and 3. The Fargo Activity Timesampling Survey (FATS) was used to code a child's behaviour into eight mutually exclusive categories (i.e., sleeping, lying down, sitting, crawling, climbing, standing, walking, and running) and three intensity levels for each category (i.e., minimal, moderate, and extreme). Summary scores were created for percentage of time in light, moderate, and vigorous activities (see Klesges and Klesges, 1987 for a detailed account of summary scores) (Klesges and Klesges, 1987).

The findings for percentage of outdoor time spent at each intensity level (i.e., MVPA, TPA, and sedentary) measured via accelerometry were pooled in R (version 3.4.0) (R Core Team, 2017) using meta-analytic techniques. Where studies used orthogonal samples within the same manuscript (i.e., Nicaise et al., 2012), weighted means and standard deviations were created across samples (except cohorts that were subdivided by age) before meta-analytically combining the results. In instances where multiple cut-points were applied to data in an individual study (i.e., Kahan et al., 2013), we opted to include the cut-points that were used most frequently in other studies (Sirard et al., 2005) to avoid over-representing the study within the meta-analysis. To aggregate the effect size estimates across studies, a linear random effects model without moderators was fit to the data (Viechtbauer, 2010). Heterogeneity was estimated using a restricted maximum-likelihood estimator, and the statistical significance of residual heterogeneity was carried out using Cochran's Q-test (Viechtbauer, 2010). Moderation analyses for the effect of age and device on percentage of outdoor time spent in MVPA, TPA, and sedentary pursuits were considered in three separate meta-analytic calculations.

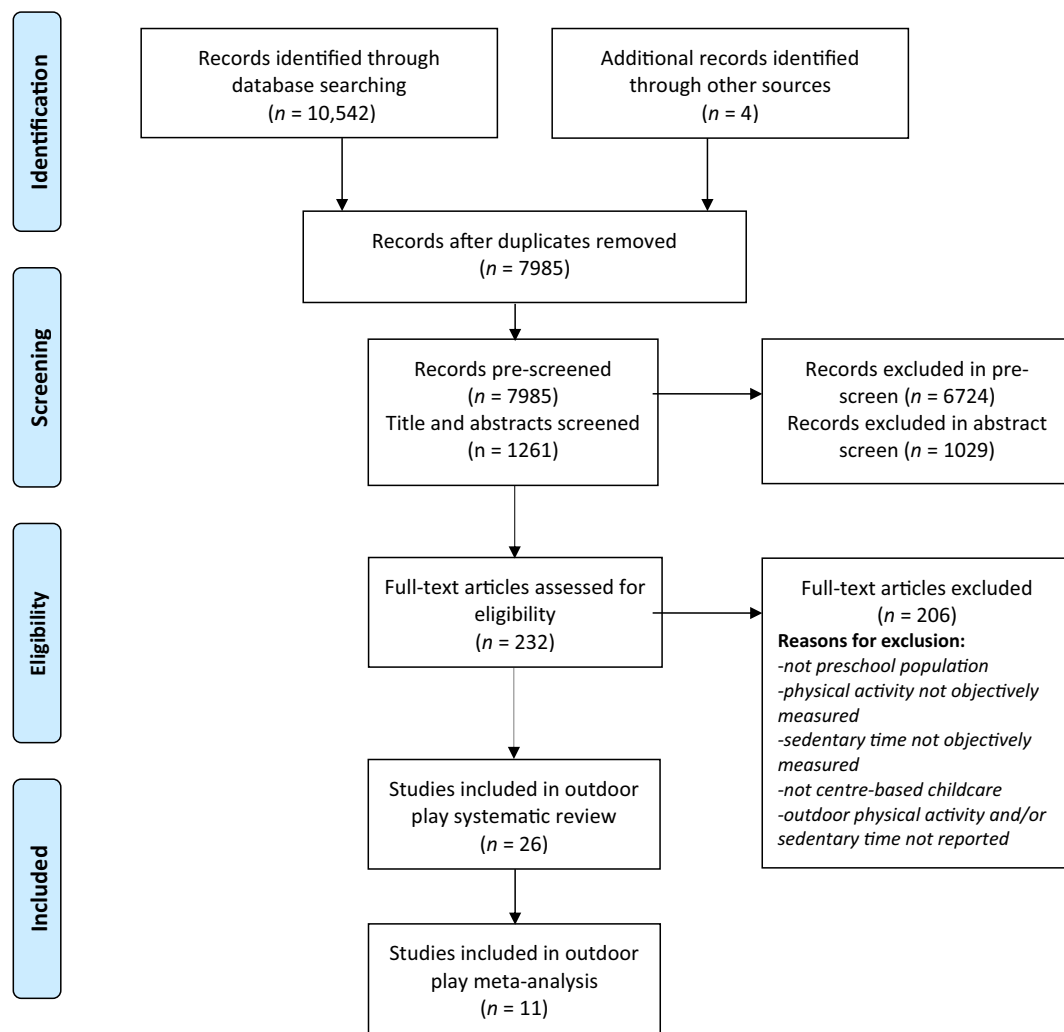


Fig. 1. Flow diagram representing the identification, screening, eligibility, and inclusion of studies in this review and meta-analysis.

3. Results

3.1. Database searches

The database search yielded 10,542 articles, while 4 additional articles were retrieved after reviewing the reference lists of included studies. No studies were found in the 'ahead of print' sections of the three online journals. After removing duplicates ($n = 2561$), reviewers pre-screened 7985 articles to remove any papers that did not focus on healthy young children. Articles underwent title and abstract screening ($n = 1261$) and full-text screening ($n = 232$), with 26 studies meeting the inclusion criteria. Articles were excluded because they did not assess physical activity and/or sedentary time in centre-based childcare or because they did not measure physical activity and/or sedentary time outdoors ($n = 206$). Eleven studies were included in the meta-analysis. See Fig. 1 for details on the identification, screening, eligibility, and inclusion process.

3.2. Study characteristics of included articles

Of the 26 studies included in this review, 18 were conducted in the United States (Brown et al., 2009a; Nicaise et al., 2011; Nicaise et al., 2012; Trost et al., 2008; Raustorp et al., 2012; Kahan et al., 2013; Becker et al., 2014; Copeland et al., 2016; Hannon and Brown, 2008; Palmer et al., 2017; Schlechter et al., 2017; Schuna Jr., 2012; Tandon et al., 2015; Brown et al., 2009b; Dowda et al., 2004; Howie et al.,

2012; Sallis et al., 1988; Sharma et al., 2011), two in Australia (Sugiyama et al., 2012; Dymment and Coleman, 2012), Canada (Vanderloo et al., 2013; Berg, 2015), and Finland (Soini et al., 2014; Reunamo et al., 2012), and one in Belgium (Van Cauwenberghe et al., 2012), Sweden (Raustorp et al., 2012), and the Netherlands (Gubbels et al., 2011) (note that one study was conducted in both Sweden and the United States) (Raustorp et al., 2012). The sample size of the studies ranged from 5 (Brown et al., 2009a) to 388 (Copeland et al., 2016) children, with a mean sample size of 109. The majority of the articles ($n = 25$) were published in 2008 or later, highlighting the infancy of this field. All 26 articles reported on physical activity, while only 18 reported on sedentary time (Nicaise et al., 2011; Nicaise et al., 2012; Raustorp et al., 2012; Vanderloo et al., 2013; Kahan et al., 2013; Copeland et al., 2016; Hannon and Brown, 2008; Schlechter et al., 2017; Schuna Jr., 2012; Tandon et al., 2015; Brown et al., 2009b; Sallis et al., 1988; Sugiyama et al., 2012; Dymment and Coleman, 2012; Berg, 2015; Soini et al., 2014; Van Cauwenberghe et al., 2012; Gubbels et al., 2011).

Twenty-two of the articles were deemed high quality (i.e., score from 7 to 10), while four scored medium quality (i.e., score from 4 to 6). Two groups of measurement tools were employed: accelerometers ($n = 11$) (Raustorp et al., 2012; Vanderloo et al., 2013; Becker et al., 2014; Copeland et al., 2016; Hannon and Brown, 2008; Palmer et al., 2017; Schlechter et al., 2017; Schuna Jr., 2012; Tandon et al., 2015; Sugiyama et al., 2012; Van Cauwenberghe et al., 2012) and direct observation ($n = 13$) (Brown et al., 2009a; Nicaise et al., 2011; Trost

Table 2
Summary of included studies that measured physical activity and sedentary time by accelerometry (*n* = 13).

Authors (year)	Country	Study design	Sample & age (yrs)	Accelerometer model	Average wear-time	Monitoring time	Epoch length	Cut-points	Physical activity (% time)	Sedentary time (% time)	Study quality (0–10) ^a
Becker et al. (2014)	U.S.	Cross-sectional	<i>N</i> = 51, Age = 4.82 (<i>SD</i> = 0.4)	ActiGraph GT1M	30 min	30 min outdoor play session	15-s	PA	MVPA = 27.23 ^Δ	Not reported	9
Copeland et al. (2016)	U.S.	Cross-sectional	<i>N</i> = 388, Age = 4.3 (<i>SD</i> = 0.7)	Actical	1 h (<i>SD</i> = 0.04) outdoor monitoring	24-h monitoring period; arrival/departure times from childcare centre recorded	15-s	PF	MVPA = 12.17 ^Δ TPA = 63.5 ^Δ	36.5 ^Δ	9
Hannon and Brown (2008)	U.S.	Repeated measures	<i>N</i> = 64, Age = 3–5	ActiGraph GT1M	15–20 min	5 days of measurement, 15–20-min outdoor free play periods	15-s	SI	MVPA = 12.17 ^Δ TPA = 42.81 ^Δ	57.64 ^Δ	8
Kahan et al. (2013) [†]	U.S.	Validation	<i>N</i> = 57, Age = 4–5	ActiGraph GT3X	Not reported	Outdoor play period (at least 15 min of data to be included)	15-s	EV, PA, SI, VC	MVPA for each set of cut-points: EV = 20.2 PA = 30.1 SI = 9.6 VC = 19.6 MVPA = 16.3 (<i>SD</i> = 9.2) TPA = 45.6 MVPA = 10.0	ST for each set of cut-points: EV = 21.2 PA = 24.3 SI = 65.6 VC = 66.1 ST = 53.8 (<i>SD</i> = 15.0)	8
Nicaise et al. (2012) [†]	U.S.	Cross-sectional	<i>N</i> = 50, Age = 4.4 (<i>SD</i> = 0.5)	ActiGraph GT3X	23.3 min (<i>SD</i> = 3.8)	Accelerometers worn during the school day on 3 occasions	15-s	SI			8
Palmer et al. (2017)	U.S.	Cross-sectional	<i>N</i> = 87, Age = 4.7 (<i>SD</i> = 0.55)	Actical	30 min	Accelerometers worn from 8:30 a.m. – 12:30 p.m. for 4 days, including a 30-min outdoor period	5-s	Not reported		Not reported	4
Raustorp et al. (2012)	Sweden and U.S.	Cross-sectional	<i>N</i> = 50, Age = 4.33 (<i>SD</i> = 0.5)	ActiGraph GT1M	U.S. = 92 min/day outdoor Sweden = 211 min/day outdoor	5 consecutive weekdays during preschool hours only	15-s	SI	MVPA: U.S. = 7.8 (<i>SD</i> = 6.1) Sweden = 6.7 (<i>SD</i> = 5.3) TPA: U.S. = 25.8 Sweden = 23.2 TPA = 34.82	U.S. = 74.0 (<i>SD</i> = 12.4) Sweden = 76.8 (<i>SD</i> = 7.1)	9
Schlechter et al. (2017)	U.S.	Cross-sectional	<i>N</i> = 73, Age = 4.36 (<i>SD</i> = 0.85)	ActiGraph GT1M	Not reported	15 days of video-recording and accelerometer measurement during preschool hours	15-s	VC		65.02	8
Schuna Jr. (2012)	U.S.	Cross-sectional	<i>N</i> = 59, Age = 4.27 (<i>SD</i> = 0.68)	ActiGraph (ActiTrainer model)	Mean fall outdoor wear-time: 56.17 min/day Mean winter outdoor wear-time: 23.12 min/day	Accelerometers worn during childcare hours only over 3 weeks at each time point (fall and winter); start and end time of outdoor play periods recorded	5-s	VC	MVPA: Fall = 20.15 ^Δ Winter = 7.35 ^Δ TPA: Fall = 31.73 ^Δ Winter = 29.24 ^Δ MVPA = 11.81 ^Δ TPA = 34.21 ^Δ	Fall = 65.21* Winter = 63.37 ^Δ	8
Sugiyama et al. (2012)	Australia	Cross-sectional	<i>N</i> = 89, Age = 4.1 (<i>SD</i> = 0.6)	ActiGraph GT1M	114.3 (<i>SD</i> = 55.0) min/day outdoor time	3 days during childcare hours only; outdoor time recorded	15-s	SI		65.79*	10
Tandon et al. (2015)	U.S.	Cross-sectional	<i>N</i> = 98, Age = 4.5	ActiGraph GT3X	34.5 min/day of outdoor measurement	Accelerometer measurement during preschool hours only (8 am–5 pm) over 4 full days at each centre	15-s	PA	MVPA = 43 ^Δ TPA = 60.5 ^Δ	40 ^Δ	8

(continued on next page)

Table 2 (continued)

Authors (year)	Country	Study design	Sample & age (yrs)	Accelerometer model	Average wear-time	Monitoring time	Epoch length	Cut-points	Physical activity (% time)	Sedentary time (% time)	Study quality (0–10) ^a
Van Cauwenberghe et al. (2012)	Belgium	Repeated measures (pilot study)	During recess: N = 128 Age = 4–6	Actical	23 (SD = 8) min/day of outdoor recess (baseline)	5 consecutive weekdays during preschool hours only; recorded outdoor recess time	15-s	VC	MVPA = 20.9 TPA = 34.8	65.3	7
Vanderloo et al. (2013)	Canada	Cross-sectional	N = 31 Age = 4.10 (SD = 0.85)	Actical	55 to 180 min of outdoor measurement	One day during childcare hours only; minimum of 4 h of wear-time to be included in analysis	15-s	PF	MVPA = 8.33 ^Δ TPA = 52.8 ^Δ	42.22 ^Δ	9

SD, standard deviation; MVPA, moderate-to-vigorous-intensity physical activity; TPA, total physical activity; ST, sedentary time; EV, Evenson; PA, Pate; PF, Pfeiffer; SI, Sirard; VC, Van Cauwenberghe. MVPA, TPA, ST, and SD were only presented in table if they appeared in original manuscript.

^Δ Conversion from min/h to % of time.

^τ Paper measured outdoor activity by accelerometry and observation.

^a Study quality based on a modified version of the Downs and Black Checklist (Downs and Black, 1998).

et al., 2008; Brown et al., 2009b; Dowda et al., 2004; Howie et al., 2012; Sallis et al., 1988; Sharma et al., 2011; Dymont and Coleman, 2012; Berg, 2015; Soini et al., 2014; Reunamo et al., 2012; Gubbels et al., 2011). Two studies measured physical activity and sedentary time outdoors via both accelerometry and direct observation (Nicaise et al., 2012; Kahan et al., 2013). Study characteristics and quality scores can be found in Tables 2 (accelerometry) and 3 (observational).

For those studies which adopted accelerometry, two different models were used: ActiGraph™ ($n = 9$) (Nicaise et al., 2011; Raustorp et al., 2012; Kahan et al., 2013; Becker et al., 2014; Hannon and Brown, 2008; Schlechter et al., 2017; Schuna Jr., 2012; Tandon et al., 2015; Sugiyama et al., 2012), and Actical™ ($n = 4$) (Vanderloo et al., 2013; Copeland et al., 2016; Palmer et al., 2017; Van Cauwenberghe et al., 2012). Outdoor accelerometer wear-time ranged from 15 (Hannon and Brown, 2008) to 211 min (Raustorp et al., 2012). All articles except for two (Palmer et al., 2017; Schuna Jr., 2012) used a 15-s epoch length. A variety of cut-points were applied within the 13 articles.

Fifteen observational studies assessed young children's physical activity levels and sedentary time outdoors at centre-based childcare. Of these, the majority used OSRAC-P ($n = 11$) (Brown et al., 2009a; Nicaise et al., 2011; Nicaise et al., 2012; Trost et al., 2008; Kahan et al., 2013; Brown et al., 2009b; Dowda et al., 2004; Howie et al., 2012; Dymont and Coleman, 2012; Soini et al., 2014; Gubbels et al., 2011), followed by SOPLAY ($n = 1$) (Berg, 2015), SOFIT-P ($n = 1$) (Sharma et al., 2011), FATS ($n = 1$) (Sallis et al., 1988), while one study did not indicate its observational measurement tool (Reunamo et al., 2012).

3.3. Characteristics of outdoor time

There was a large degree of variability in the outdoor playtime reported in the 26 studies. Outdoor time measurements ranged from a minimum of 10 min (Dymont and Coleman, 2012) to a maximum of 211 min (Raustorp et al., 2012), with a mean of 45.2 min, over a typical 7.5-h day at childcare. Common recording periods for accelerometry and direct observation were 20 to 30 min outdoors, for 4 consecutive days.

3.4. Percentage of outdoor time engaged in physical activity and sedentary time

3.4.1. Accelerometer

Young children spent between 6.7% (Raustorp et al., 2012) to 43% (Tandon et al., 2015) of their outdoor time engaged in MVPA ($n = 12$), with a mean of 17.8%. TPA ($n = 11$) ranged from 23.2% (Raustorp et al., 2012) to 63.5% (Copeland et al., 2016) of outdoor play time, with a mean of 42.4%. Sedentary time accounted for 21.2% (Kahan et al., 2013) to 76.8% (Raustorp et al., 2012) of young children's outdoor sessions at centre-based childcare, with a mean of 55.2%. Mean percentage of time spent in MVPA according to specific cut-points were 33.4% for Pate et al. (2010), 20.2% for Evenson et al. (2008), 19.9% for Van Cauwenberghe et al. (n.d.), 10.2% for Sirard et al. (2005), and 9.6% for Pfeiffer et al. (2006) Refer to Table 2 for specific percentages of activity for individual studies. Studies conducted in the United States reported mean percentage of time in MVPA, TPA, and sedentary time as 12.17%, 43.36% and 55.89%, respectively; compared to studies from other countries (i.e., Canada, Australia, Finland, Belgium, Sweden, and the Netherlands), which collectively averaged MVPA, TPA, and sedentary percentages of 11.94%, 36.25%, and 62.53% respectively.

3.4.2. Direct observation

The most common presentation of results for the observational studies was percentage of time spent in MVPA ($n = 15$), and percentage of time engaged in sedentary behaviours ($n = 9$). No studies reported TPA using this measurement method, as each tool lacked the sensitivity necessary to differentiate sedentary from light, and light from moderate activity. Among the observational studies, MVPA varied between 2%

Table 3
Summary of included studies that measured physical activity and sedentary time by direct observation ($n = 15$).

Authors (year)	Country	Study design	Sample N & age (yrs)	Tool	Length of observation	Number of observations	Physical activity (% time)	Sedentary time (% time)	Study quality (0–10) ^a
Berg (2015)	Canada	Cross-sectional	N = 161, Age = 3–5	SOPLAY	Not reported	2268	Moderate activities = 29.3 Vigorous activities = 20 MVPA = 49.3 ^Δ MVPA = 18.4	50.6	6
Brown et al. (2009a)	U.S.	Cross-sectional	N = 5, Age = 4	OSRAC-P	30-min outdoor observation 3 or 4 days per week; 5-s to observe, 25-s to record (2 observations/min)	Not reported		Not reported	7
Brown et al. (2009b)	U.S.	Cross-sectional	N = 372, Age = 3–5	OSRAC-P	30-min observation period; focal child, momentary time sampling (5-s observe, 25-s record; 2 observations per minute for 60 intervals per session)	Total observed intervals outside = 29,685	LPA = 27 MVPA = 17	56	8
Dowda et al. (2004)	U.S.	Cross-sectional	N = 266, Age = 3–5	OSRAP	1 h on 2 or 3 days; activity of a selected child observed for 15-s	Not reported	MVPA = 27.7	Not reported	10
Dymond and Coleman (2012)	Australia	Cross-sectional	N = 16, Age = 4–5	OSRAC-P	Each child observed for 50-min over a 30-day period with a max of 10-mins of observation per day; data collected in 1-min intervals (50-s observation, 10-s recording)	Not reported	MPA = 30 VPA = 23.9	46.1	6
Gubbels et al. (2011)	Netherlands	Cross-sectional	N = 175, Age = 2–3	OSRAC-P	Momentary time sampling with observations lasting 15-s, with 30-s following for recording; procedure repeated 4 times over 3-min for each child; each child observed for twice	2800	MVPA = 21.3	31.2	10
Howie et al. (2012)	U.S.	Cross-sectional	N = 231, Age = 3–5	OSRAC-P	5-s observation followed by 25-s of recording for each 30-s interval; each child observed for 10–12 20-min sessions	600–720 30-s intervals per child	Intervals observed running = 25.2 ^Δ	Not reported	8
Kahan et al. (2013) [†]	U.S.	Validation	N = 57, Age = 4–5	OSRAC-P	Outdoor recess (minimum of 15-min to be included)	Not reported	MVPA = 14.3	57.5	8
Nicaise et al. (2011)	U.S.	Cross-sectional	N = 51, Age = 4.45 (SD = 0.52)	OSRAC-P	15- to 30-min outdoor play periods; each child observed on two occasions	204	LPA = 26.4–30.7 MVPA = 12.4–20.7 Mean MVPA = 16.56 ^Δ MVPA = 5.5 (SD = 5.1) LPA = 14.3 (SD = 9.3) TPA = 19.8 ^Δ	48.6–61.2	8
Nicaise et al. (2012) [†]	U.S.	Cross-sectional	N = 50, Age = 4–5	OSRAC-P	Avg. observation time = 23.3 mins (SD = 3.8, range 15–30 min); 5 s observation/25-s recording intervals; each child was observed twice	214		80.2 (SD = 8.4)	8
Reunamo et al. (2012)	Finland	Cross-sectional	N = 62 daycare centres	Observation	8 a.m.–12 p.m. observed during outdoor time	19,606	LPA = 56.4 ^Δ MPA = 33.3 ^Δ VPA = 10.3 ^Δ	Not reported	4
Sallis et al. (1988)	U.S.	Cross-sectional	N = 33, Age = 3.9 (SD = 0.7)	FATS	30-min unstructured free-play session on playground during two consecutive days; 10-s observe-record time sampling procedures	Not reported	MPA = 31 (SD = 13) VPA = 11 (SD = 8)	58 (SD = 13)	9
Sharma et al. (2011)	U.S.	Validation	Phase II: N = 27, Age = 3.84 (SD = 0.52)	SOFIT-P	PA level coded 30-s using momentary time sampling (20-s observation, 10-s recording); 4 outdoor observations on one school day; measuring “active times” only	10 observations per child outdoors	MVPA = 45.2	Not reported	8

(continued on next page)

Table 3 (continued)

Authors (year)	Country	Study design	Sample N & age (yrs)	Tool	Length of observation	Number of observations	Physical activity (% time)	Sedentary time (% time)	Study quality (0–10) ^a
Soini et al. (2014)	Finland	Cross-sectional	N = 81, Age = 3	OSRAC-P (modified version)	15-s observation followed by 30-s recording; repeated 8 times over 6-min for each child; each child observed at least twice per day over 3 consecutive days	1978 observations × 8 = 15,824 single observation intervals	MVPA = 2	46	9
Trost et al. (2008)	U.S.	Intervention	Control group: N = 42, Age = 4 (SD = 0.7)	OSRAP	Control class preschoolers observed on 16.1 occasions (range 10–20) with each session lasting on average 14–15 min (28.5 30-s epochs), 14.9% of observations outdoors	10,114 30-s observations (control); % of 30-s observation epochs outdoors in MVPA = 25%		Not reported	9

SOPALY, System for Observing Play and Leisure Activity in Youth; MVPA, moderate-to vigorous-intensity physical activity; ST, sedentary time; OSRAC-P, Observational System for Recording Physical Activity in Children–Preschool; LPA, light physical activity; SD, standard deviation; FATS, Fargo Activity Timesampling Survey; SOFIT-P, System for Observing Fitness Instruction Time. MVPA, TPA, ST, and SD were only presented in table if they appeared in original text.

^a Conversion from min/h to % of time.

[†] Paper measured outdoor activity by accelerometry and observation.

^a Study quality based on a modified version of the Downs and Black Checklist (Downs and Black, 1998).

Table 4

Pooled results without moderators for percentage of time in MVPA, TPA and sedentary activity.

	Total n	No. of studies	Estimate	SE	95% CI	I ²
% Time in MVPA	900	11	13.17	1.37	10.49 to 15.85	68.36%
% Time in TPA	776	9	40.29	4.00	32.45 to 48.13	94.27%
% Sedentary time	922	11	56.86	3.34	50.31 to 63.4	90.03%

SE, standard error; CI, confidence interval; MVPA, moderate-to vigorous-intensity physical activity; TPA, total physical activity.

(Soini et al., 2014) and 49.3% (Berg, 2015), with a mean of 27.1%, while sedentary time ranged from 46.1% (Dymont and Coleman, 2012) to 80.2% (Nicaise et al., 2012), with a mean of 53.4%. Specific percentages of activity for individual studies are presented in Table 3.

3.4.3. Meta-analyses

Of the 13 studies which measured physical activity and sedentary time during outdoor play periods at childcare with accelerometers, 11 provided sufficient data for inclusion in meta-analyses ($n_{MVPA} = 10$; (Nicaise et al., 2012; Raustorp et al., 2012; Vanderloo et al., 2013; Kahan et al., 2013; Becker et al., 2014; Copeland et al., 2016; Hannon and Brown, 2008; Schuna Jr., 2012; Sugiyama et al., 2012; Van Cauwenberghe et al., 2012) $n_{TPA} = 8$; (Vanderloo et al., 2014; Nicaise et al., 2012; Raustorp et al., 2012; Copeland et al., 2016; Hannon and Brown, 2008; Schlechter et al., 2017; Schuna Jr., 2012; Van Cauwenberghe et al., 2012) $n_{Sedentary\ Time} = 10$) (Vanderloo et al., 2014). Table 4 provides the pooled results without moderators for percentage of time engaged in MVPA, TPA and sedentary pursuits.

Device (i.e., type of accelerometer) and age were explored as moderators of MVPA, TPA and sedentary time. Device was found to be a significant moderator for percentage of time spent in TPA [$Q(1) = 5.41, p = 0.02$], but was not statistically significant for percentage time in MVPA [$Q(1) = 0.06, p = 0.80$] or percentage of time spent in sedentary activities [$Q(1) = 2.22, p = 0.14$]. Studies using Actical devices were found to have a larger percentage of time spent in TPA (mean = 53.99%, 95% CI: 34.09% to 73.89%) as compared with ActiGraph devices (mean = 31.65%, 95% CI: 22.59% to 40.70%). Age was found to be a significant moderator for percentage of time spent in MVPA [$Q(1) = 4.32, p = 0.04$], demonstrating a significant increase with age. Neither percentage of time spent in TPA [$Q(1) = 0.51, p = 0.48$] nor percentage of time spent in sedentary activities [$Q(1) = 0.86, p = 0.35$] showed statistically significant moderating effects of age.

4. Discussion

The childcare setting has received considerable attention as a venue to support physical activity among young children (Finn et al., 2002). Moreover, outdoor playtime has been widely recognized as important for supporting physical activity among young children (Tremblay et al., 2015). In light of the influx of evidence in this field, this systematic review and meta-analysis aimed to synthesize the physical activity levels and sedentary time of young children during outdoor play periods at centre-based childcare. Many young children spend a substantial portion of their day enrolled in these venues, which have been recognized as being influential on young children's activity behaviours. As such, garnering a better understanding of their outdoor activity, where there is prime opportunity for unstructured free play and gross motor movement, was necessary.

4.1. Young children's outdoor physical activity levels

The first key finding from this review is that young children are more active while outdoors at centre-based childcare (averaged 40% of time spent in TPA as evidenced in this systematic review and 43.77% via meta-analysis), compared to indoor activity rates, where young children have been reported to engage in 20% of their time in TPA (Vanderloo et al., 2013). Although variability did exist among the accelerometer-based studies (i.e., heterogeneity was substantive across studies included in the meta-analysis), young children spent a large portion of their time engaged in MVPA (2% to 53.9%) and TPA (23.2% to 63.5%). To ease comparability, if mean percentages were applied to a 1-hour outdoor play period at childcare, young children would average 10.7 min in MVPA and 25.5 min/h in TPA. In Ontario, the *Child Care and Early Years Act* stipulates that for every 6 h in care, young children in centre-based childcare are required to be provided with a minimum of two, 1-hour play periods outside (Government of Ontario, 2017). This rate of outdoor physical activity during care hours would contribute to more than a quarter of the daily TPA recommendation of 180 min for this population (Tremblay et al., 2017), which is accepted globally (i.e., Canada, Australia, and United Kingdom) (Tremblay et al., 2017; Okely et al., 2017; Department of Health, 2011). However, it may be difficult for young children to accumulate enough physical activity to achieve this benchmark during the remainder of their waking hours. Young children have been noted to be substantially less active indoors compared to outdoors when they are at childcare (Raustorp et al., 2012; Vanderloo et al., 2013), and may not have as many active opportunities when they return home due to busy schedules, meal time, and bedtime routines. For example, McKenzie et al. (1992) reported that preschoolers were significantly more active at childcare during recess than they were when they returned home for the evening (McKenzie et al., 1992). Given the large proportion of young children's waking hours are spent in the childcare environment (Cleveland et al., 2008), and the consistent observation of higher activity levels outdoors than indoors, offering more opportunities for outdoor active play during care hours could increase the likelihood of this population achieving, or even surpassing, the recommended levels of daily physical activity (Tremblay et al., 2017; Department of Health, 2011, 2014).

While studies in this review came from a variety of countries, no apparent geographical variability in physical activity levels and sedentary time were observed. While discrepancies in physical activity and sedentary time is possible between countries, given the varied climates and seasonal differences (Tucker and Gilliland, 2007), or region-specific childcare policies (Ward et al., 2009; Vanderloo and Tucker, 2017; Copeland et al., 2012), both noted to influence physical activity rates, no differences were identified in the current review.

The studies using direct observation to assess young children's outdoor physical activity levels at childcare reported higher values of MVPA than the accelerometer-based studies. If the average percentage of time spent in MVPA were applied to a 1-hour window, young children in the observational studies would spend an additional 7.21 min in MVPA compared to those in the accelerometer-based studies. These results are consistent with the literature regarding observational measures tending to overestimate MVPA compared to accelerometers (Pate et al., 2010). The 5-point scales used for many of the observational studies only classified sedentary behaviour (levels 1 and 2) and MVPA (levels 4 and 5).

4.2. Differences in measurement tools

Due to the complex nature of measuring physical activity, many different tools have been used to assess these behaviours in this young population. As evidenced in the current review, much variability exists in the type of tools for activity measurement (i.e., accelerometry-based vs. observational studies), but also within the individual measurement tools utilized (i.e., different cut-points, epoch length, device models,

length of observation). Although the observational studies, on average, identified young children as more active than when measured via accelerometry, accelerometry is currently considered the gold standard for measuring the activity levels of this population as these devices are sensitive to young children's sporadic movement patterns (Hills et al., 2014). While young children are typically recognized as an active cohort, their activity is rarely done for a continuous period of time, highlighting the difficulty in quantifying their movement (Cliff et al., 2009). It is important to be mindful when interpreting results of a single study, as different devices, cut-points, wear-time/observation periods, and observational tools yield distinctive results (Vanderloo et al., 2016), and it may not always be appropriate to make direct comparisons.

In this review, wide variance was observed in the physical activity levels accumulated by participants measured by different accelerometer devices. More specifically, studies that used Actical™ accelerometers recorded a higher percentage of time in TPA (53.99%) compared to those that utilized ActiGraph™ accelerometers (31.65%), while MVPA and sedentary time were not significantly different. Comparing the two accelerometers by having children wear both devices side-by-side on their hips, Vanderloo and colleagues found the opposite result; Actical™ accelerometers reported significantly higher levels of sedentary time, while the ActiGraph™ device captured significantly more physical activity (both TPA and MVPA) (Vanderloo et al., 2016). Disparity between individual studies in this review could be a result of the different models of ActiGraph™ (i.e., uniaxial vs triaxial) utilized, or could also be a consequence of the short window of activity (in a location which is conducive to more physical activity) that was observed in the present review, in comparison to the full day of observation conducted in Vanderloo et al.'s study.

Further evidence is required for this cohort in relation to short monitoring time, as reliability of estimates of TPA are maximized with 7 days of monitoring for 10 h per day; however, ≥ 3 h of monitoring on ≥ 3 days has also provided reliable estimates (Penpraze et al., 2006). Variability as a consequence of the specific cut-points applied to the raw physical activity data has also been identified as impacting how the activity is classified into different intensities. For example, Kahan et al. (2013) found that depending on the applied cut-points (Evanson et al., 2008; Van Cauwenberghe et al., 2011; Sirard et al., 2005; Pfeiffer et al., 2006), the percentage of time spent in MVPA varied from 9.6% to 30.1%. It is important to note that some cut-points are age-dependent (i.e., Sirard et al., 2005) and others are not, leaving potential to over/underestimate activity levels (Cliff et al., 2009). Fluctuating epoch lengths also pose a challenge when measuring physical activity via accelerometry. Although Hislop et al. (2012) indicated that using shorter epoch lengths of 15-s resulted in significantly higher number of minutes of MVPA compared to a 60-s epoch length (Hislop et al., 2012), the two studies in this review which used a 5-s epoch presented comparable levels of MVPA to the remaining studies which utilized 15-s epochs.

Despite the noted challenges of measuring physical activity using accelerometry, it is still widely utilized in the field. Moving forward, it may be beneficial to combine use of accelerometers with an observational component, as adopting both forms of measurement in combination would provide a strong picture of activity intensities via accelerometry, while observational tools provide important contextual information regarding type and location of activity behaviour. Further, the development of a feasible, valid, and a reliable measure of outdoor play will complement such effort.

4.3. Young children's outdoor sedentary time

Although it was found that young children in centre-based childcare were quite active outdoors, high rates of sedentary time were also observed (~55% as per systematic review). Prolonged periods of sedentary time are associated with non-communicable disease, and mortality

(Goel, 2015; Barton, 2012), increasing the concern over the large amount of time this populations spends in this behaviour. If percentages of time spent in sedentary time from the accelerometer-based and observational studies in this review were applied to a 1-hour timeframe, young children would average 27.74 and 32.04 min being sedentary outdoors, respectively. While acknowledging young children's sedentary time is intermittent in nature, this finding is still important to consider in light of the Canadian, Australian, and UK guidelines which state that this population should not engage in more than an hour consecutively in sedentary behaviours such as sitting or being restrained in a stroller or high chair, and should limit screen-viewing to less than 1 h per day (Tremblay et al., 2017; Okely et al., 2017; Committee on Public Education, 2001). Though this proportion of time spent being sedentary outdoors is high, Vanderloo et al. (2013) concluded that preschoolers spent twice the amount of sedentary time indoors compared to outdoors. Therefore, the outdoor environment still appears more conducive to lowering young children's sedentary time at childcare. Intervention studies aimed at decreasing sedentary time in childcare have shown promise (Nicaise et al., 2012; Hannon and Brown, 2008; Van Cauwenberghe et al., 2012).

4.4. Position statement on active outdoor play

In 2015, a position statement on active outdoor play was released. Derived from expert consensus and extensive review of the literature, this statement highlighted the need for children to get outdoors and be active, encouraging children to engage in risky play and to be adventurous in their environments (Tremblay et al., 2015). Nicaise et al. (2012) support that preschoolers should be allowed to explore their surroundings and be provided with opportunities to interact with natural elements on the playground. Elements of nature present children with new and exciting obstacles to overcome (e.g., climbing a tree, running on uneven surfaces, etc.), while also giving them the freedom to play in an unstructured manner, which builds confidence, imagination, autonomy, and problem solving skills (Tremblay et al., 2015). Young children should be afforded more outdoor playtime in environments which are conducive to increased physical activity. Despite the growing literature supporting the need for children to be provided with more active play opportunities outdoors following provincial safety regulations (Tremblay et al., 2015; Tandon et al., 2015; Sugiyama et al., 2012; Tucker et al., 2017), policies within childcare centres are lacking in regard to outdoor playtime (Story et al., 2006). It is clear from this review that intervention and policy development efforts are warranted to be more consistent with recent literature supporting the importance of active outdoor play for young children during childcare hours.

4.5. Limitations

Firstly, as the majority of research on this topic is extremely recent, all study designs and qualities of evidence were included to provide a comprehensive overview of this body of literature. While the overall strength of this paper may be affected by the inclusion of all qualities of evidence, it is important to note that most of the included studies ($n = 22$ out of 26) were of high quality. Secondly, different cut-points were used to classify activity intensity, and various accelerometer models were used to measure physical activity and sedentary time, challenging the comparability between studies. Thirdly, activity intensity was derived from validated cut-points but no postural assessment was conducted to complement the sedentary time estimates. Fourthly, some studies had to be excluded from this review because we were unable to interpret or discern exact physical activity and/or sedentary time from the figures provided, and some authors were not responsive when additional information was requested. Fifthly, the wear-time/observation periods for the included studies were not equivalent. When necessary, simple calculations were performed to convert minutes of MVPA, TPA, and sedentary time per hour into

percentage of time so results could be compared between studies. Extrapolating results from the included studies with short outdoor play periods remains difficult as applying percentages of time in different activity intensities over extended periods suggests that young children are sustaining the same rate of activity for the duration of an outdoor play session, which may not be accurate. Sixthly, environmental factors such as play equipment and playground density were not considered in this review, which have been previously recognized as influencing activity behaviours. Finally, not all studies provided sufficient information to be included in the meta-analysis.

5. Conclusion

This is the first systematic review and meta-analysis to assess young children's levels of physical activity and sedentary time during outdoor play sessions at centre-based childcare. While young children were found to be more active outdoors at centre-based childcare (13% of time in MVPA and 40% in TPA as per meta-analyses), this population still engaged in a considerable amount of sedentary time outdoors (upwards of 60%). The results from this review emphasize the need for future research involving interventions to increase young children's physical activity levels and decrease their sedentary time while outdoors at childcare. As many children spend a large percentage of their day in this environment, it is an ideal setting to establish healthy physical activity habits and reduce sedentary time. Increasing the duration and/or frequency of outdoor time (Greever et al., 2015; Copeland et al., 2016), and altering the outdoor play environment (e.g., addition of portable play equipment, incorporation of natural elements into playground area, etc.) (Nicaise et al., 2012; Palmer et al., 2017), may prove effective at increasing and supporting young children's physical activity levels in childcare during periods of outdoor play.

Acknowledgements

The authors would like to thank Marisa Tippet (Research and Instructional Services Librarian, University of Western Ontario) for her assistance in developing the current review's search strategy. Truelove and Bruijns were supported by the Ontario Graduate Scholarship (2016–2017); Vanderloo was supported by the CIHR Frederick Banting and Charles Best Doctoral Research Award (2014–2016); and Tucker was supported by an Early Research Award from the Ontario Ministry of Research and Innovation.

Conflicts of interest

The authors declare no conflicts of interest.

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Update

Preventive Medicine

Volume 127, Issue , October 2019, Page

DOI: <https://doi.org/10.1016/j.ypmed.2019.105814>



Corrigendum

Corrigendum to “Physical activity and sedentary time during childcare outdoor play sessions: A systematic review and meta-analysis” [Pre. Med. 108 (2018) 74–85]



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The authors regret that, due to a minor analytic error, some estimates presented in the original manuscript are incorrect. However, the corrections do not imply changes in interpretation.

In the abstract, the text should read “Meta-analysis of 11 accelerometer studies found that children spend a mean of (95% CI) of: 13.52% (10.48–16.65) of outdoor playtime in MVPA, 40.44% (32.53–48.36) in TPA, and 56.86% (50.24–63.48) in sedentary time”.

In Section 3.4.3, the second paragraph should read as follows: “Device (i.e., type of accelerometer) and age were explored as moderators of MVPA, TPA, and sedentary time. Moderation by device approached significance for TPA [$Q(1) = 3.13$, $p = 0.08$], but was not statistically significant for percentage time in MVPA [$Q(1) = 0.01$, $p = 0.91$] or percentage of time spent in sedentary activities [$Q(1) = 2.52$, $p = 0.11$]. Studies using Actical devices were found to have a larger percentage of time spent in TPA (mean = 50.34%, 95% CI: 33.71% to 66.97%) as compared with ActiGraph devices (mean = 36.23%, 95% CI: 28.67% to 43.79%). Age was not found to be a significant moderator for percentage of time spent in MVPA, TPA, or sedentary activities.”

Table 4 has been updated below.

Table 4

Pooled results without moderators for percentage of time in MVPA, TPA and sedentary activity.

	Total n	No. of stu- dies	Estimate	SE	95% CI	I ²
% time in MV- PA	900	10	13.52	1.55	10.48 to 16.55	97.69%
% time in TPA	776	8	40.44	4.04	32.53 to 48.36	99.18%
% sedentary ti- me	922	10	56.86	3.38	50.24 to 63.48	98.00%

Notes. SE, standard error; CI, confidence interval; MVPA, moderate-to vigorous-intensity physical activity; TPA, total physical activity.

Similarly, in the second paragraph of Section 4.2, percentage of time in TPA for Actical accelerometers should read 50.34% (rather than 53.99%), and percentage of time in TPA for ActiGraph should read 36.23% (rather than 31.65%).

The authors would like to apologise for any inconvenience caused.

DOI of original article: <https://doi.org/10.1016/j.ypmed.2017.12.022>

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<https://doi.org/10.1016/j.ypmed.2019.105814>